July 2026

Large Science Models:

Foundation Models for Generalizable Insights Into Complex Systems with Psycho-social Application

DARPA-EA-25-02-05-MAGICS-PA-025
HR0011-26-3-E016

Zero Knowledge Discovery

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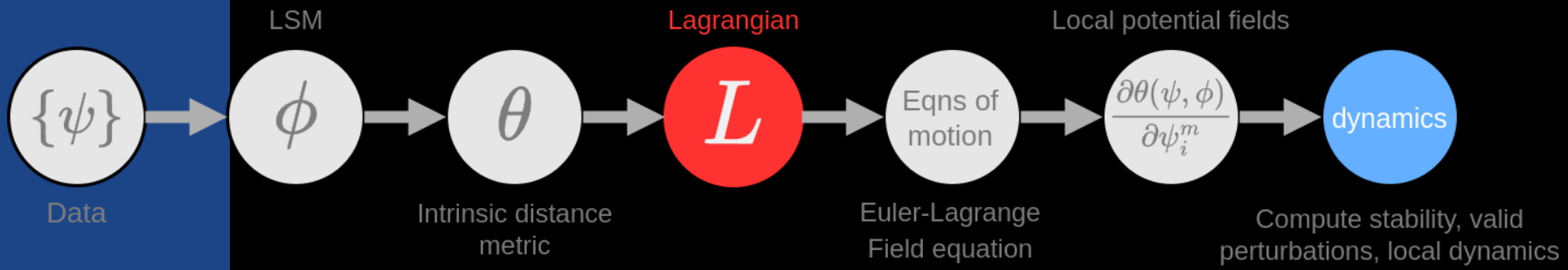
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DARPA



Proposed Concept



- Develop Foundation models of complex systems with
  - *hundreds to thousands of evolving variables with apriori unknown cross-talk*
  - *no governing equations are know a priori*
  - *reflexivity: system changes if observed*
- Learn intrinsic system geometry from data
- Derive equations of motion with variational principles (*stationary action on Lagrangian*).
- Inference under data sparsity
- Detect data (in)sufficiency, adapt to model drift
- Support forward simulation and perturbation analysis
- Digital twins of individuals & groups wrt to opinion dynamics

MAGICS Alignment

Data inference boundaries & limitations

Component LSM predictors enforce statistical significance of splits in recursive partitioning, ensuring precise uncertainty quantification

Alignment validation

Precise validation protocols to assess process drift triggering re-calibration/training

Complex phenomena

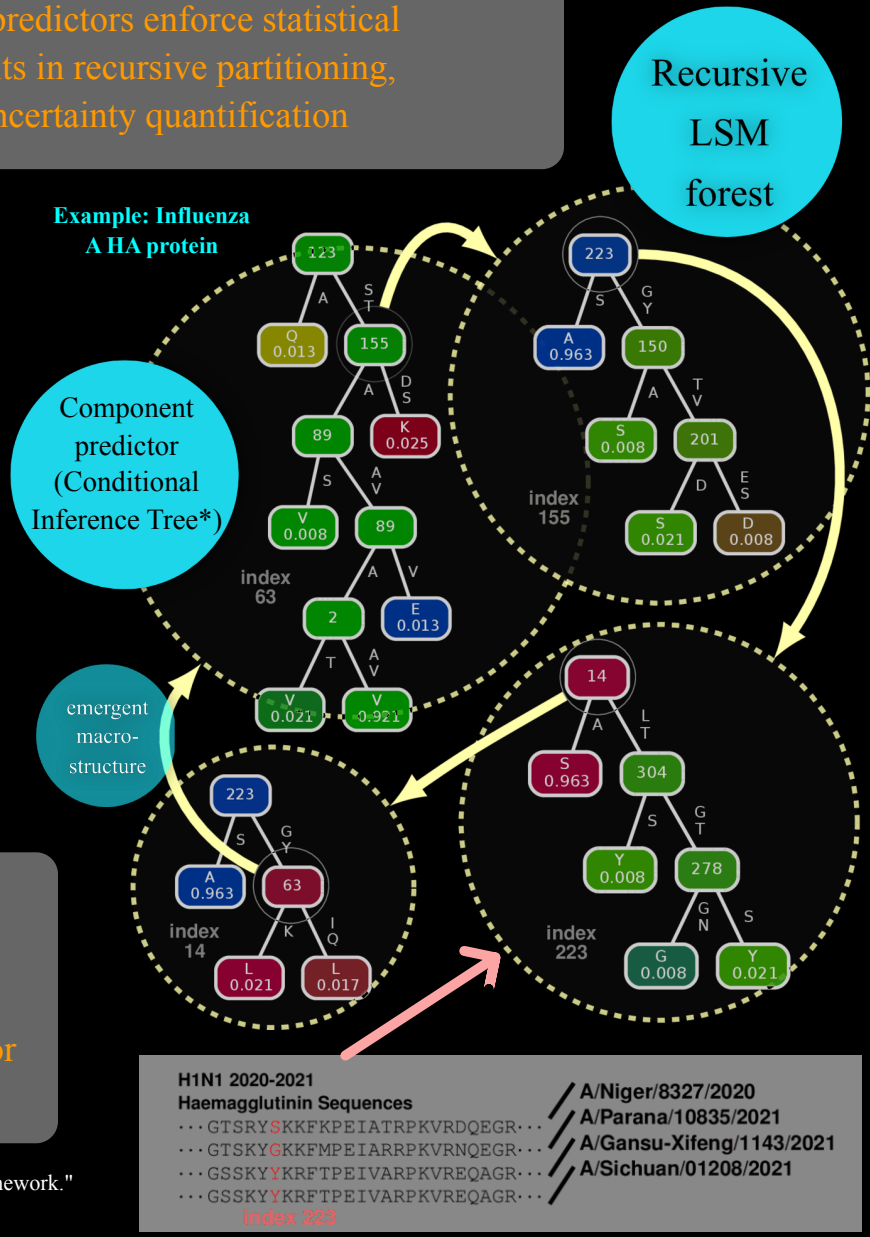
Scalable to thousands to millions of variables, intrinsic reflexivity

Adaptation to model obsolence

Built-in flexibility for changing contexts and non-ergodicity

Psychosocial domain limitations

Validate social theories with granular simulations from digital twins of opinion dynamics and social behavior



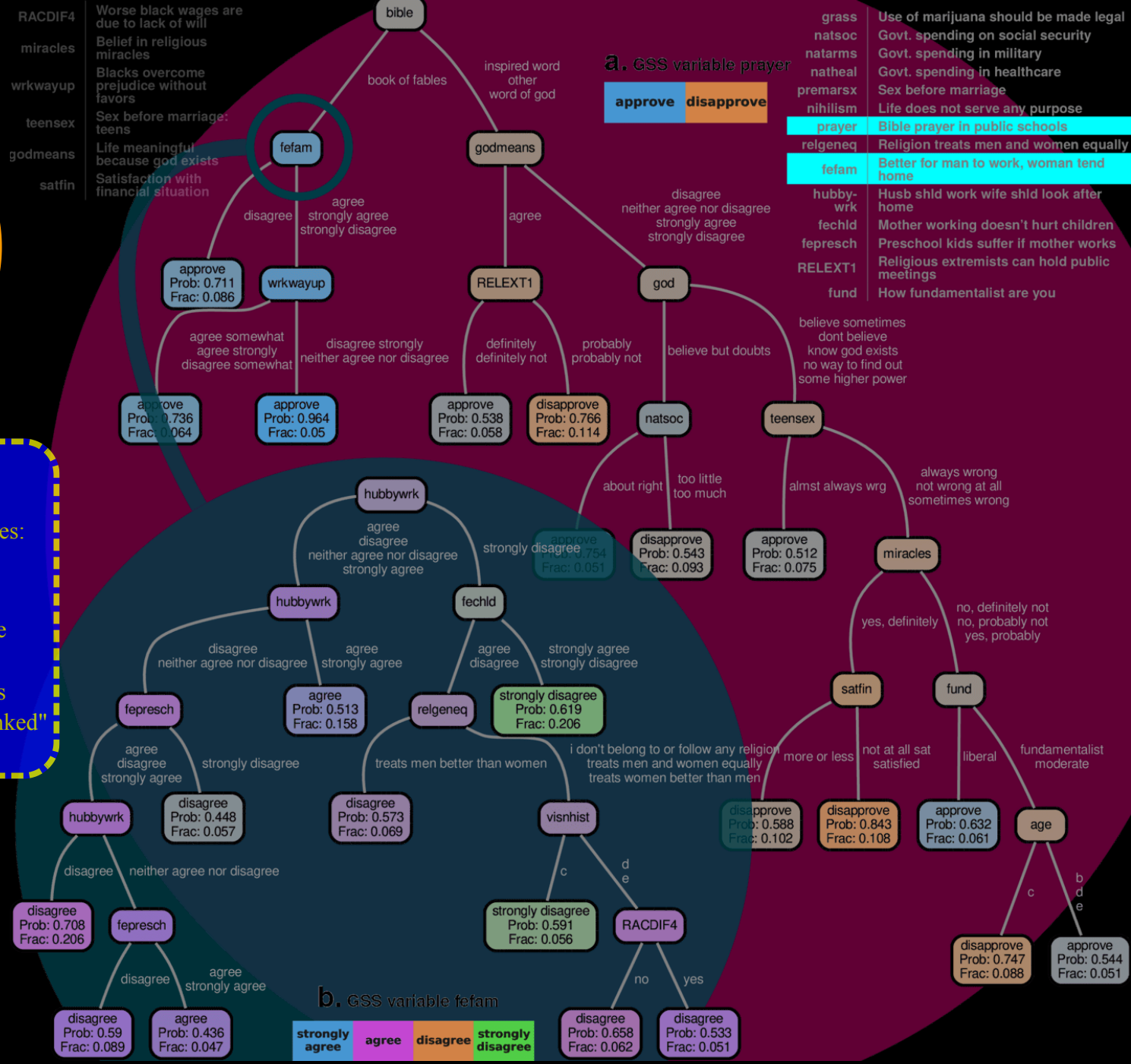
*Hothorn, Torsten, Kurt Hornik, and Achim Zeileis. "Unbiased recursive partitioning: A conditional inference framework." Journal of Computational and Graphical statistics 15, no. 3 (2006): 651-674.

LSM Forest

GSS 2018 dataset

Recursive LSM forest: hyperlinked nodes capturing emergent macro-structures

- Set of ***conditional inference trees*** (CIT)
  - Strict statistical guarantees: quantifies inference uncertainty
- Each tree models exactly one variable as a function of potentially all other variables
- Non-leaf nodes are "hyperlinked" to other trees



Large Science Models

Computationally tractable LSM tree structure given, as proposed, hundreds to thousands of observable variables.

https://paraknowledge.ai/data/trees2018/obey.svg

GSS 2018 dataset

- Each predictor is inferred independently
- Can scale up to thousands of variables in Python implementation
- Further scale-up $10^6 - 10^8$ needs C/C++ implementation

Full Example of Hyperlinked Trees

https://paraknowledge.ai/data/trees2018/

Publications and Submissions (2026)

01 Wu, K., Li, F., & Chattopadhyay, I. (2026). Emergenet: A digital twin of influenza A evolution for vaccine strain forecast assessment. Military Medicine. https://doi.org/10.1093/milmed/usag255

02 Dahal, H., Young, D., Evans, J., & Chattopadhyay, I. (2026). The fabric of opinion: How interdependent attitudes constrain behavior, and evolve over time. Nature Communications. Submitted.

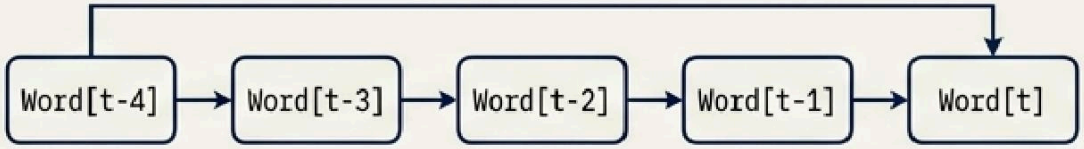
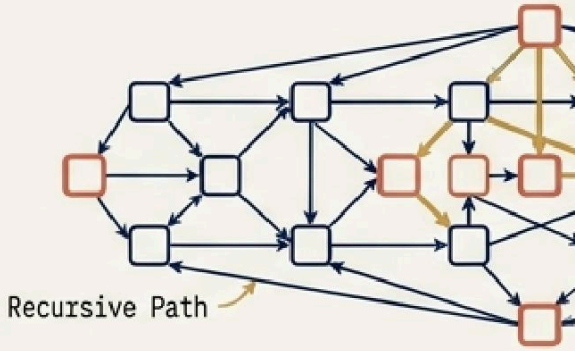
03 Dahal, H., & Chattopadhyay, I. (2026). How good is your synthetic social data? Conditional alignment and novelty across 3 to the 2026 Computational Social Science Society of the Americas Conference (CSS 2026), Santa Fe, New Mexico. Submitted.

04 Dahal, H., & Chattopadhyay, I. (2026). Distribution-identifying diagnostics for synthetic multivariate data: Fidelity, re concentration. Biostatistics. Submitted.

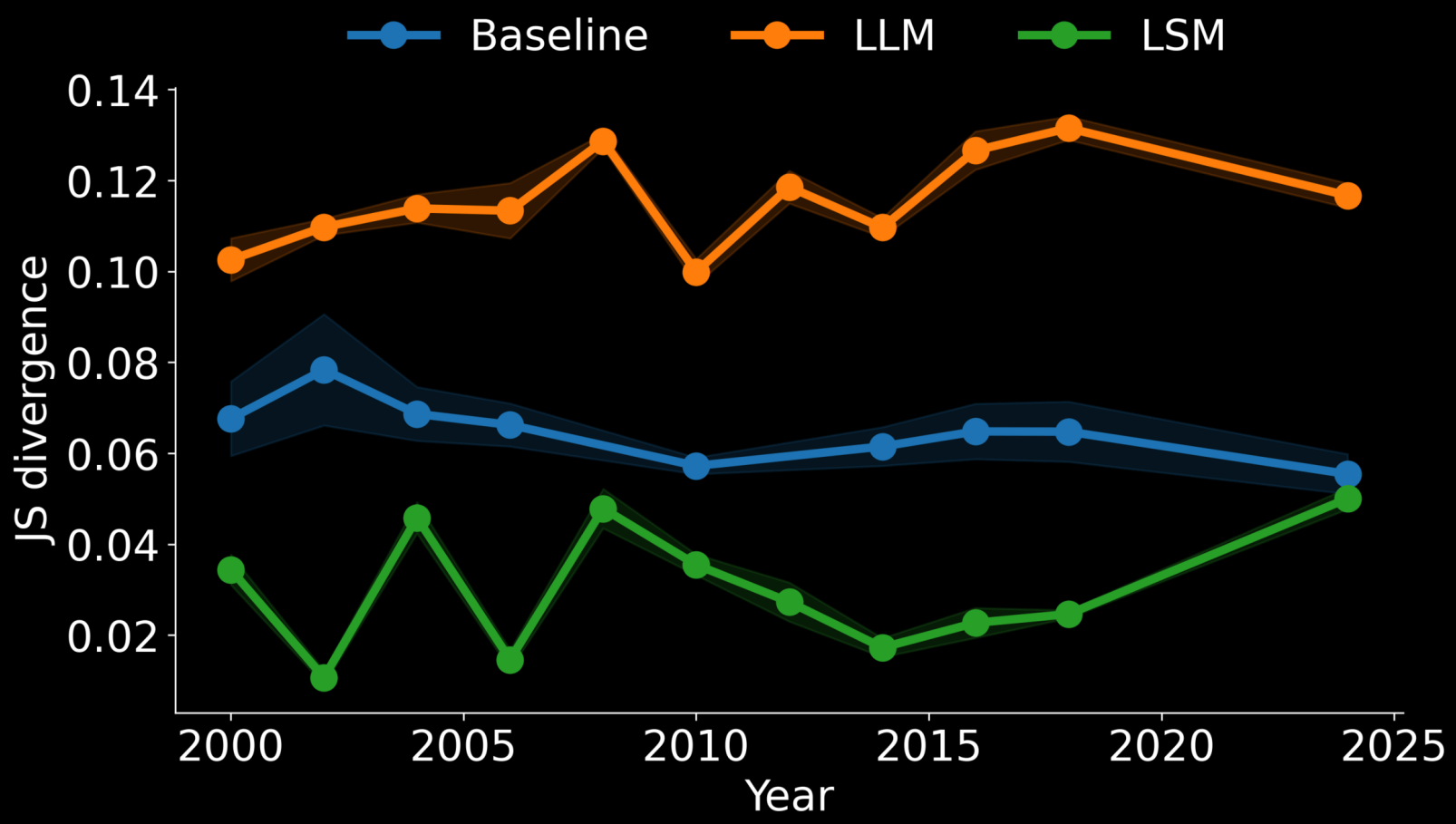
05 Cai, H., Yu, W., Lu, R., Chattopadhyay, I., Zhang, X., & Liu, J. (2026). Synthetic longitudinal tabular data generation v Submitted.

06 Dahal, H., Chattopadhyay, I., & Evans, J. (2026). Geometry of opinions: Building data-driven digital twins from social su 2026 Computational Social Science Society of the Americas Conference (CSS 2026), Santa Fe, New Mexico. Submitted.

Core Scientific Contribution: LSMs vs

Large Language Models (LLMs)	Large Science M
 *Sequential Next-Token Prediction Pipeline*	 *Recursive Conditional Inference Fores*
Core Architecture	Semantic Probabilistic Word Prediction	Latent Dependency Non-P
State Mapping	Hallucination-prone; unconstrained by empirical realities	Learns the constrained state to induce an intrinsic, d
Validation Metric	Next-token accuracy	Rigorous out-of-sample s

Key Insight: A general modeling stack now exists for high-dimen where explicit mechanistic specification is unavaila


Baseline
LLM
LSM
0.14
0.12
0.10
0.08
0.06
0.04
0.02
JS divergence
2000
2005
2010
2015
2020
2025
Year

The Empirical Foundation: Global Opinion Geometry

4.05M+
Participants



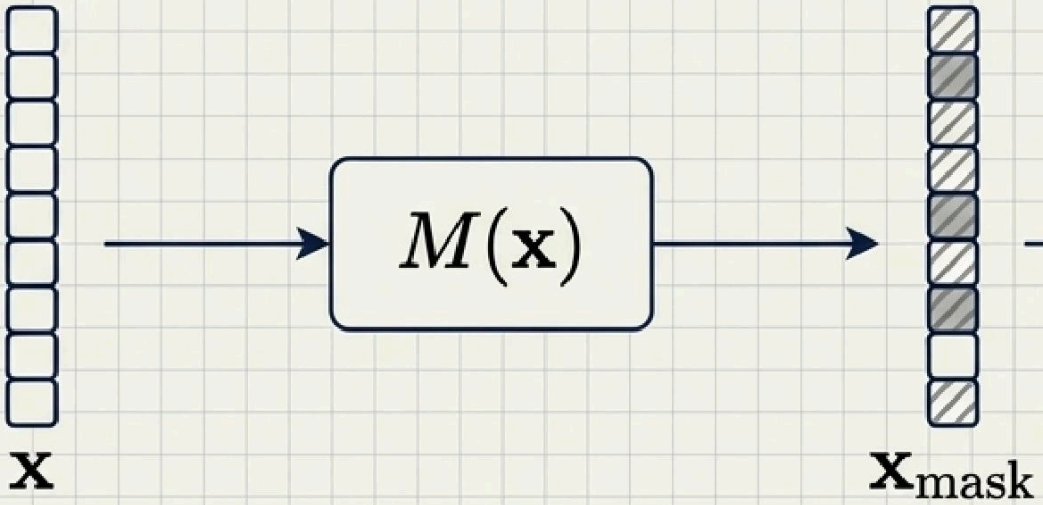
1972–2025
Coverage

- GSS
- Eurobarometer
- Afrobarometer
- WVS



Insight: These combined
empirical geometry required
"Digital Twins" of global po
mapped by time, geography,

Rigorous Validation: Masked Sample Rec



Distortion Reduction $\Delta_{\text{recon}} = 100 \times \left(1 - \frac{\epsilon}{\epsilon}\right.$

GSS Validation

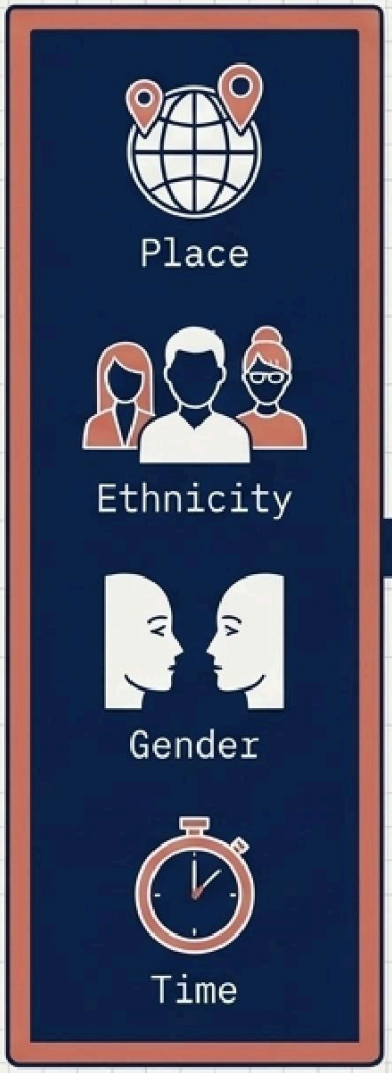
- Achieved 54.9% mean distortion reduction (mean post-reconstruction distortion fell to 45.1%).
- Over 90.1% of out-of-sample participants improved.

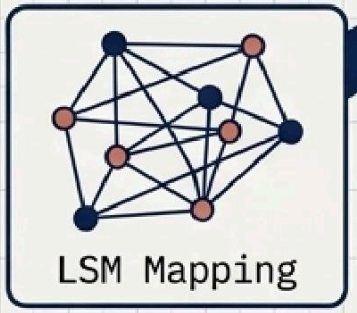
Eurobaromete

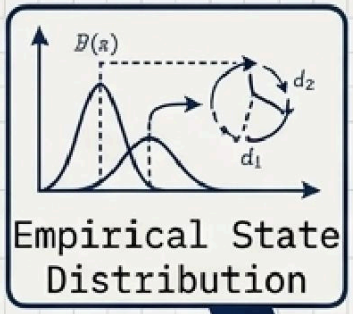
- Achieved 60.9% mean disto
- Over 99.5% of out-of-samp

Conclusion: Statistically significant predictive capability far exceeding median or ra

Architecture of DTAG (Digital Twin Anchored Gen

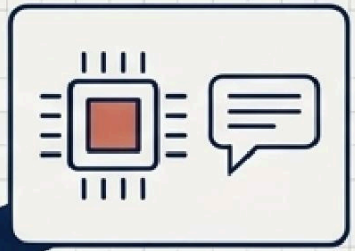






DTAG Cycle

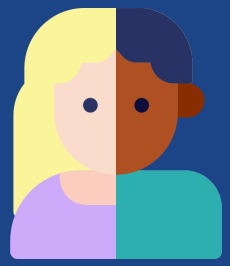


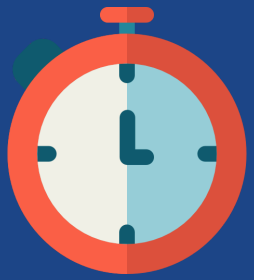


DTAG: Global Digital Twin of Opinions









Recall Bail etal.

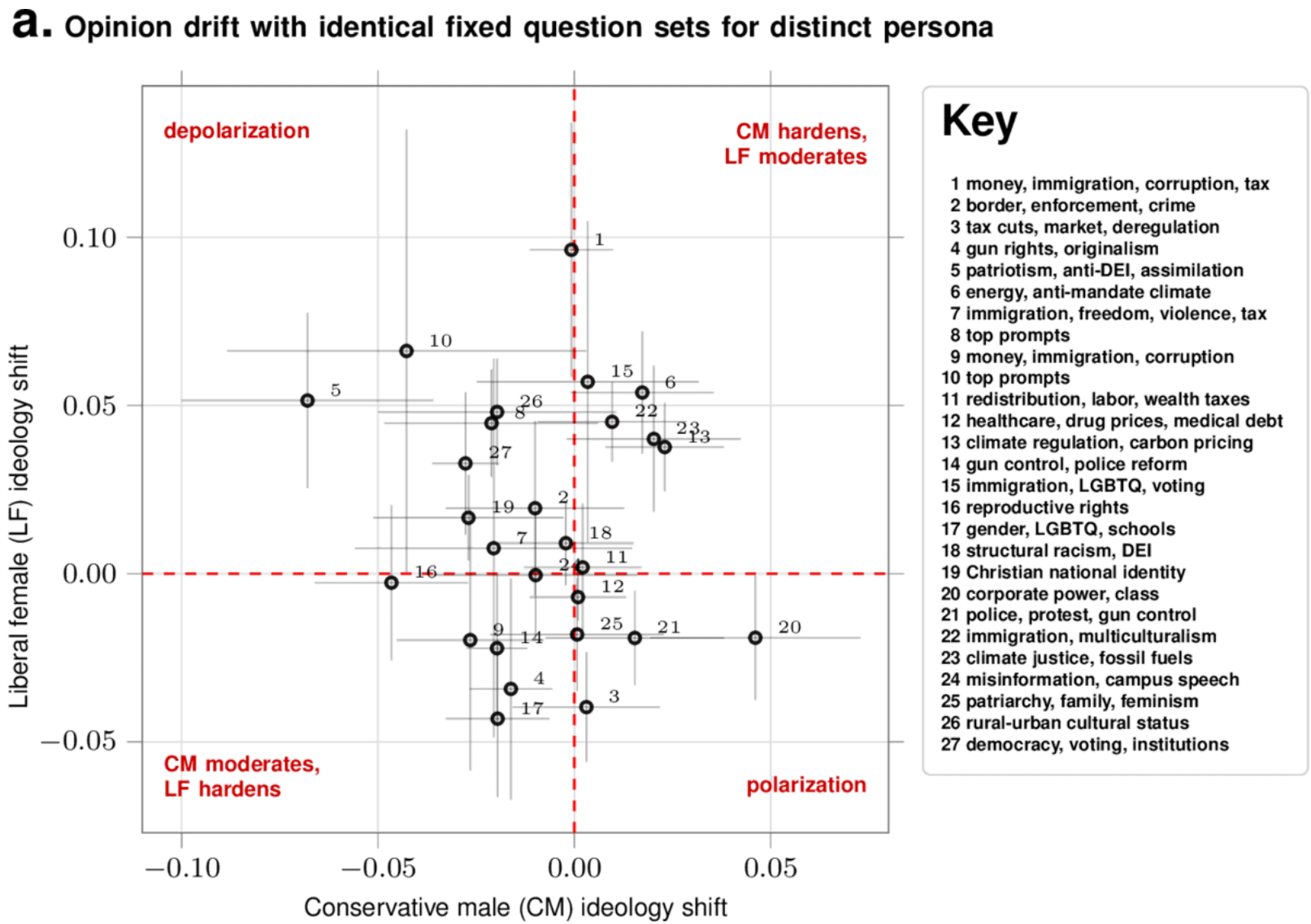
"Exposure to opposing views on social media can increase political polarization" by Christopher A. Bail et al., published in PNAS in September 2018 (Vol. 115, No. 37, pp. 9216–9221; DOI: 10.1073/pnas.1804840115)

Perturbing with opposing views made conservatives more conservative (statistically significant), liberals more liberal (not statistically significant)

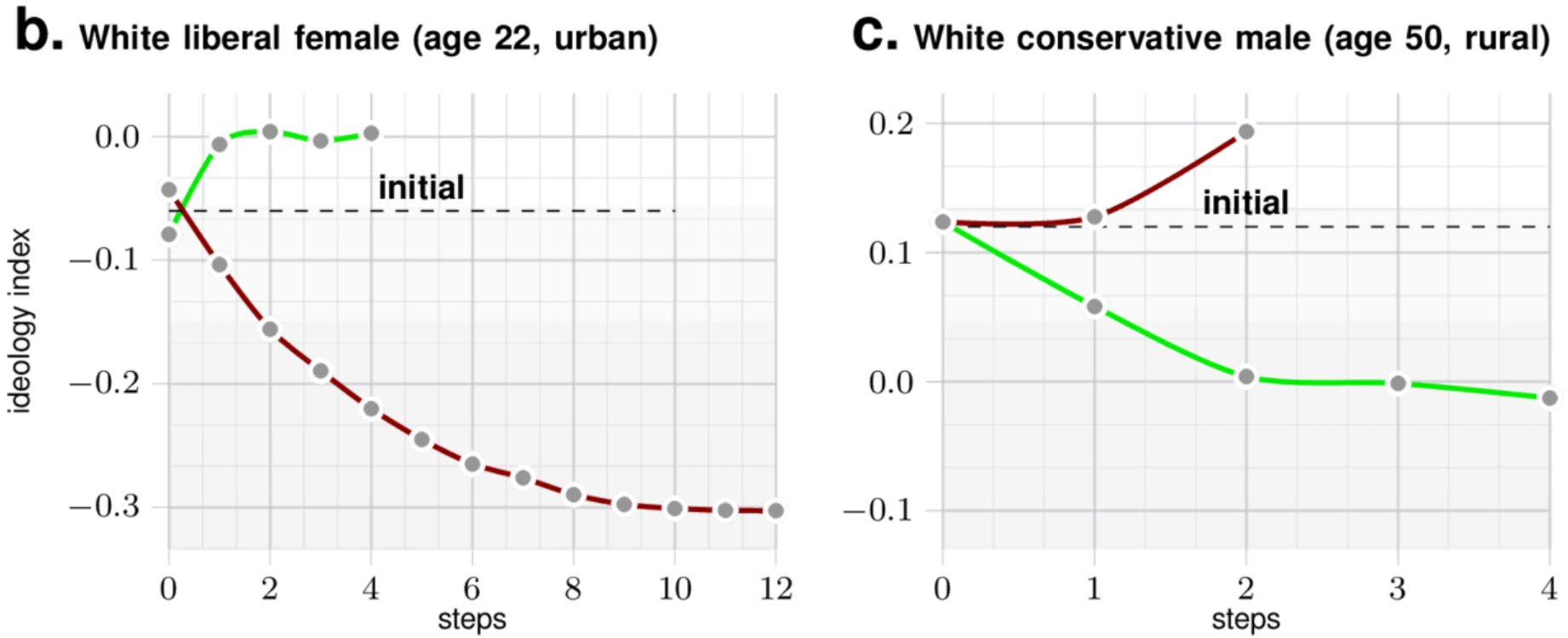
We find more general possibilities: We can make world-views go more extreme or less extreme based on the line of questions and the persona

LSM

- **Fixed Question Sets Exist That Move Different Persona Towards Polarization/Depolarization**



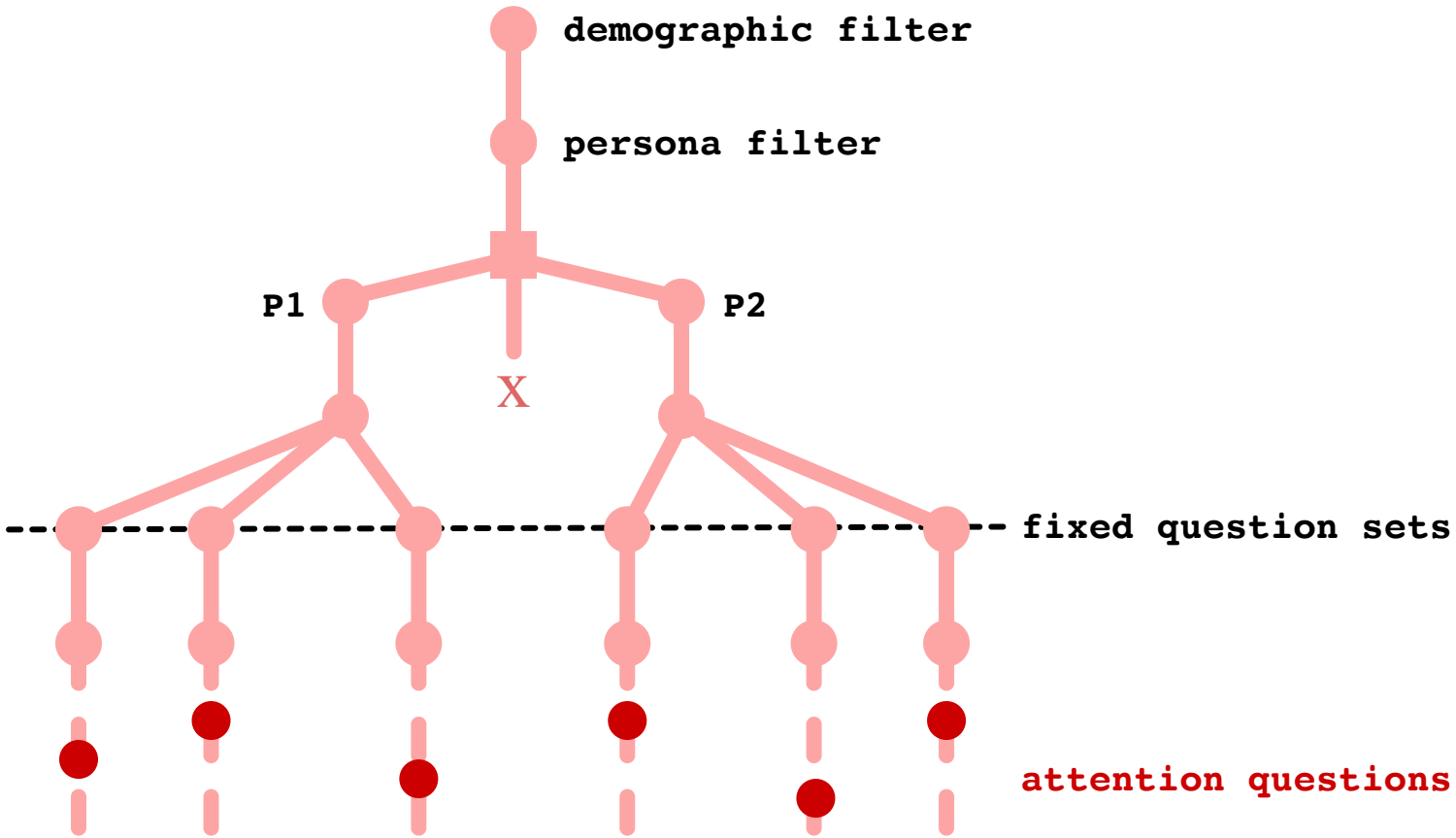
- **We can optimize question sequences to move the same persona in a chosen direction**



- **Note: It is relatively easy to polarize than to depolarize**

Prospective validation in Human Cohorts

- **Beyond MAGICS Scope**
- Prolific Experiments using non-DARPA External Funding with UKy IRB approval



Future Trajectory & Zero-Burden Cli

Upcoming Theoretical Milestones (DARPA 5-8)

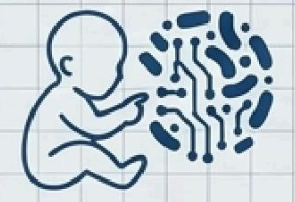
Model drift sensing validation (non-stationary drift)

Data sufficiency tracking (complexity conservation)

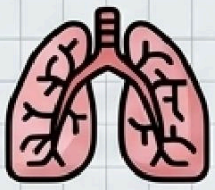
C
a
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PROP
Prosp
quest
polar

Clinical Horizons & Zero-Burden AI



Lab-test for ASD predicting with >90% AUC at 1 year (via behavioral/demographic latent mapping)



Inters
diagno
profil
from a

LSMs provide a common, universal mathematical language for pa
simulation, and prediction across social, biological, and cl